

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 35

KARST TOPOGRAPHY

- **EROSIONAL LANDFORMS**
- **DEPOSITIONAL LANDFORMS**

KARST TOPOGRAPHY

LIMESTONE AND CHALK



LIMESTONE

- 1** Limestone and chalk are sedimentary rocks of organic origin derived from the accumulation of corals and shell in the sea.
- 2** In its pure state, **limestone** is made up of calcite or calcium carbonate, but where magnesium is also present it is termed **dolomite**.
- 3** Chalk is a very pure form of limestone, white and rather soft.

KARST TOPOGRAPHY

LIMESTONE AND CHALK



CHALK

- 4** Limestone is soluble in rain-water, which with carbon dioxide from the air forms a weak acid.
- 5** A region with a large stretch of limestone therefore possess a very distinct type of topography.
- 6** It is termed as a **karst region**, a name derived from the **Karst district** of **Yugoslavia** where such topography is particularly well developed.

KARST TOPOGRAPHY



Example of Karst Landscape

KARST TOPOGRAPHY

CHARACTERISTIC FEATURE OF A KARST REGION



Rivers disappear underground in Karst topography

- 1** There is a general absence of surface drainage as most of the surface water has gone underground.
- 2** Streams rising on other rocks only flow over limestone for a short distance and then disappear underground.

KARST TOPOGRAPHY

CHARACTERISTIC FEATURE OF A KARST REGION

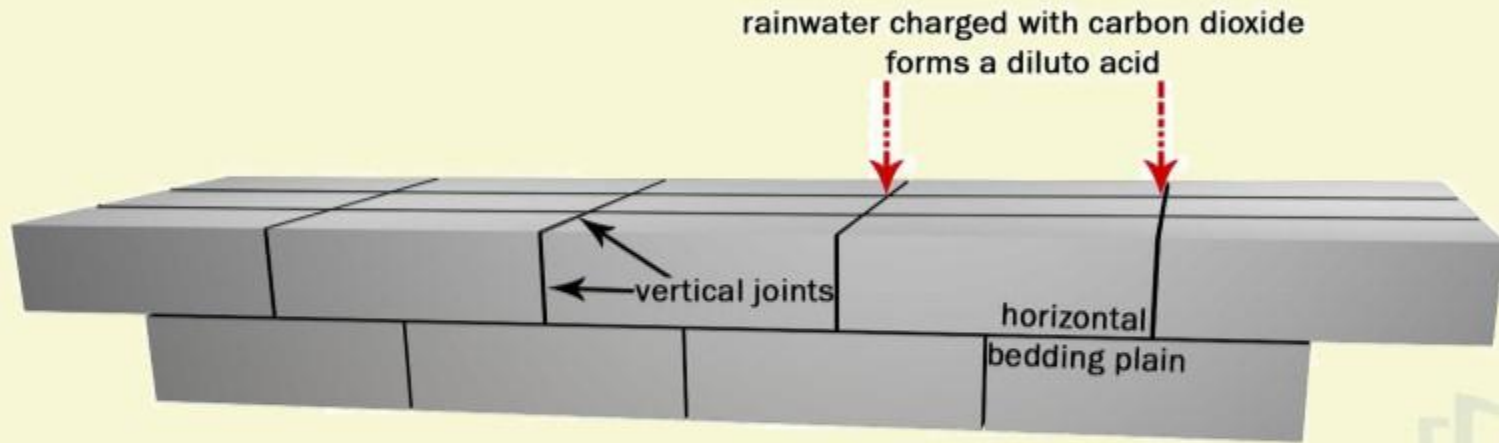


SPRING OR RESURGENCE

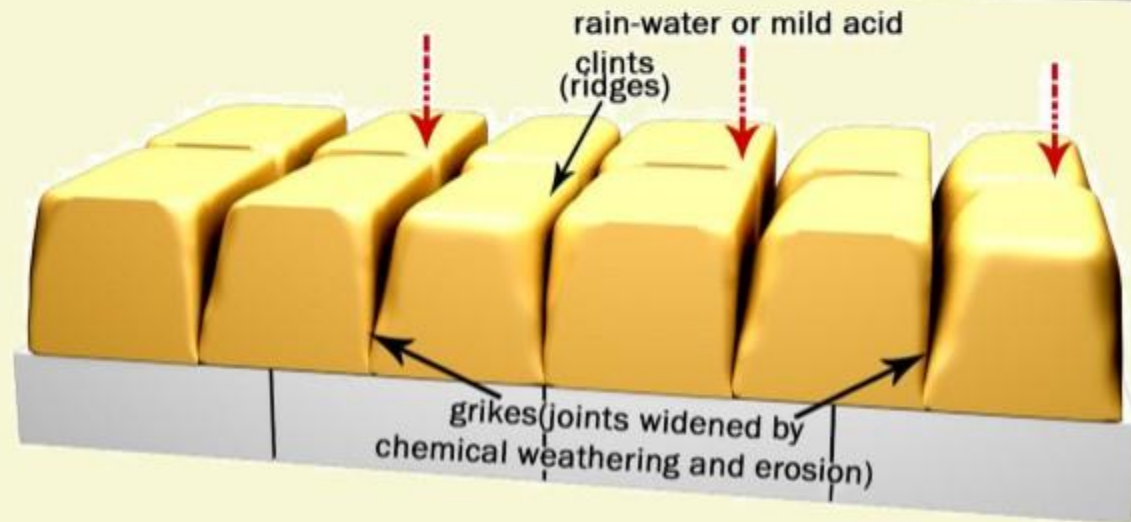
- 3** For the greater part of their course, they cut their way along the joints and fissures of the rock wearing out a system of underground channels.
- 4** When the water penetrates to the base of the limestone and meets non-porous rocks it re-emerges onto the surface as a **spring or resurgence**.

KARST TOPOGRAPHY

CHARACTERISTIC FEATURE OF A KARST REGION

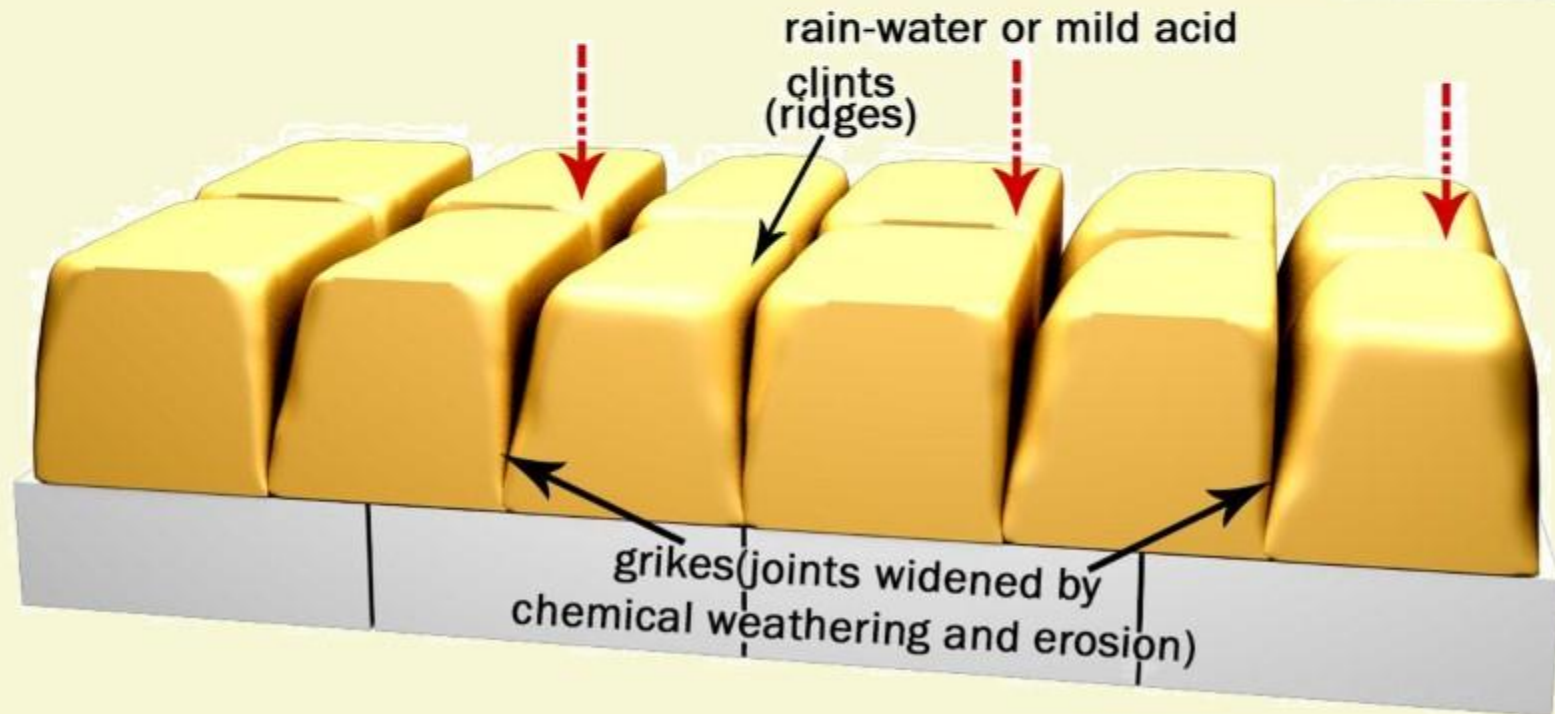
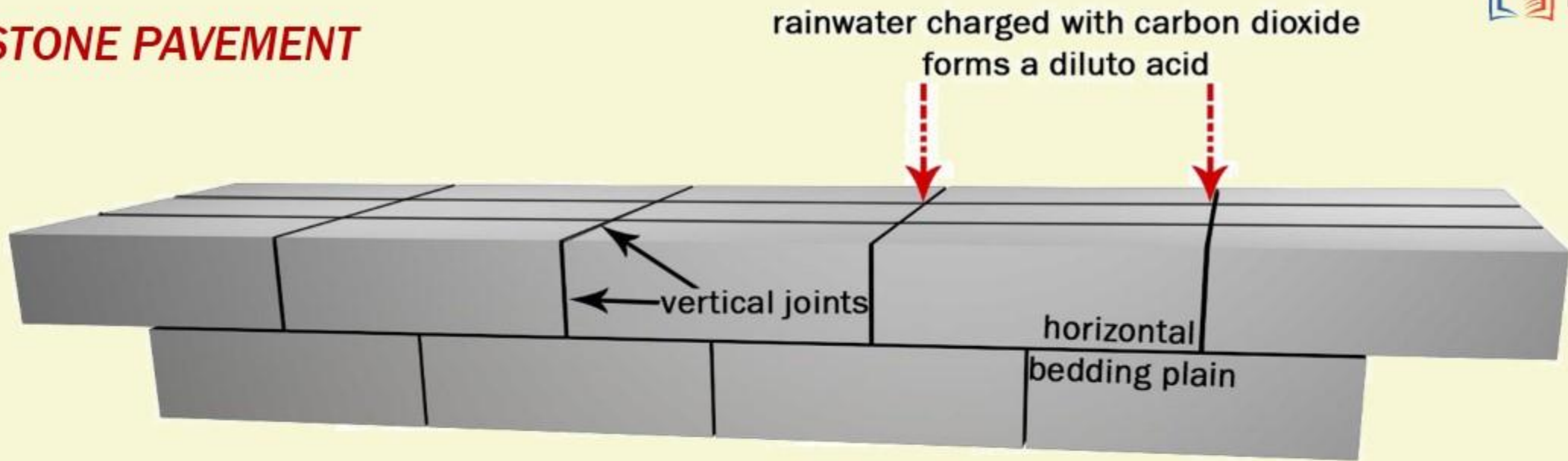


5 Limestones are well jointed and it is through these joints and cracks that rain-water finds its way into the underlying rock.



6 Progressive widening by solution enlarges these cracks into trenches and a most intriguing feature called **limestone pavement** is developed.

LIMESTONE PAVEMENT



KARST TOPOGRAPHY

GROUNDWATER



- 1 Here the interest is not on groundwater as a resource. Our focus is on the work of groundwater in the erosion of landmasses and evolution of landforms.
- 2 The surface water **percolates well when the rocks are permeable, thinly bedded and highly jointed and cracked.**
- 3 After vertically going down to some depth, the water under the ground flows horizontally through the bedding planes or joints or through the materials themselves.

KARST TOPOGRAPHY

GROUNDWATER



- 4 It is this downward and horizontal movement of water which causes the rocks to erode.
- 5 The results of the work of groundwater cannot be seen in all types of rocks.
- 6 But in rocks like limestones or dolomites rich in calcium carbonate, the surface water as well as groundwater through the chemical process of solution and precipitation deposition develop varieties of landforms.
- 7 These two process of solution and precipitation deposition are active in limestones or dolomites occurring either exclusively or interbedded with other rocks.

KARST TOPOGRAPHY

GROUNDWATER



KARST TOPOGRAPHY

8 Any limestone or dolomitic region showing typical landforms produced by the action of groundwater through the processes of **solution and deposition** is called '**Karst topography**' after the typical topography developed in limestone rocks of karst region in the Balkans adjacent to Adriatic sea.

9 The karst topography, like the fluvial and glacier topography, is also characterised by erosional and depositional landforms.

KARST TOPOGRAPHY

EROSIONAL LANDFORMS

1 SWALLOW HOLE



- 1 Small to medium sized round to sub-rounded shallow depressions called swallow holes form on the surface of limestones through solution.
- 2 That is, small depressions get carved out where rain-water sinks into the limestone pavement, forming what is known as swallow holes.
- 3 These holes grow in size continuously through solvent action and give way to sink holes.

KARST TOPOGRAPHY

EROSIONAL LANDFORMS

2

SINK HOLE



1

Sinkholes are very common in limestone/karst areas.

2

A sinkhole is an opening **more or less circular at the top and funnel-shaped** towards the bottom with sizes varying in area from a few sq.m to a hectare and with depth from a less than half a metre to thirty metres or more.

3

Quite often, **sinkholes are covered up with soil mantle and appear as shallow water pools.**

4

Anybody stepping over such pools would go down like it happens in **quicksands in deserts.**

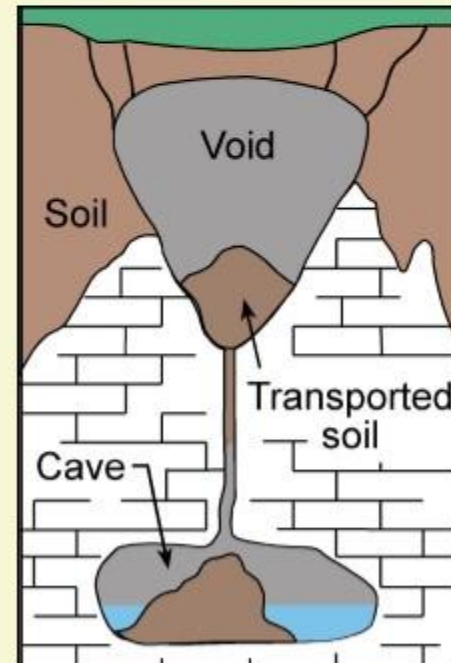
KARST TOPOGRAPHY

EROSIONAL LANDFORMS

2

SINK HOLE

- 5** Some of these form solely through solution action and others might start as solution forms first and if the bottom of a sinkhole forms the roof of a void or cave underground, it might collapse leaving a large hole opening into a cave or a void below (collapse sinks).



KARST TOPOGRAPHY

EROSIONAL LANDFORMS

3

DOLINES

- 1** Where a number of swallow holes coalesce a larger hollow is formed and is called a doline.



- 2** Sometimes the term doline is used to refer the collapse sinks.

- 3** Solution sinks are more common than collapse sinks.

- 4** Quite often the surface run-off simply goes down swallow and sink holes and flow as underground streams and re-emerge at a distance downstream through a cave opening.

- 5** Several dolines may merge as a result of subsidence to form a larger depression called an uvala.

KARST TOPOGRAPHY

EROSIONAL LANDFORMS

3

DOLINES



6 Gradually, most of the surface of the limestone is eaten away by these pits and trenches, leaving it extremely irregular with a maze of points, grooves and 'ridges or lapies'.

7 Especially, these ridges or lapies form due to differential solution activity along parallel to sub-parallel joints.

8 The lapie field may eventually turn into somewhat smooth limestone pavements.

KARST TOPOGRAPHY

EROSIONAL LANDFORMS

4

CAVES



1 In areas, where there are alternating beds of rocks (shales, sandstones, quartzites) with limestones or dolomities in between or in areas where limestones are dense, massive and occurring as thick beds, cave formation is prominent.

2 Water percolates down either through the materials or through cracks and joints and moves horizontally along bedding planes.

3 It is along these bedding planes that the limestone dissolves and long and narrow to wide gaps called caves result.

KARST TOPOGRAPHY

EROSIONAL LANDFORMS

4

CAVES



4

There can be a maze of caves at different elevations depending upon the limestone beds and intervening rocks.

5

Caves normally have an opening through which cave streams are discharged.

6

Caves having openings at both the ends are called tunnels.

KARST TOPOGRAPHY

DEPOSITIONAL LANDFORMS



- 1 Many depositional forms develop within the limestone caves.
- 2 The chief chemical in limestone is calcium carbonate which is easily soluble in carbonated water (carbon dioxide absorbed rainwater).
- 3 This calcium carbonate is deposited when the water carrying it in solution evaporates or loses its carbon dioxide as it trickles over rough rock surfaces.

KARST TOPOGRAPHY

DEPOSITIONAL LANDFORMS

STALACTITES, STALAGMITES AND PILLARS

- 1 Stalactites hang downwards as pinnacles of different diameters. Normally they are broad at their bases and taper towards the free ends showing up in variety of forms.



KARST TOPOGRAPHY

DEPOSITIONAL LANDFORMS

STALACTITES, STALAGMITES AND PILLARS



2 Stalagmites rise up from the floor of the caves. In fact, stalagmites form due to dripping water from the surface (or through the thin pipe) of the stalactite.

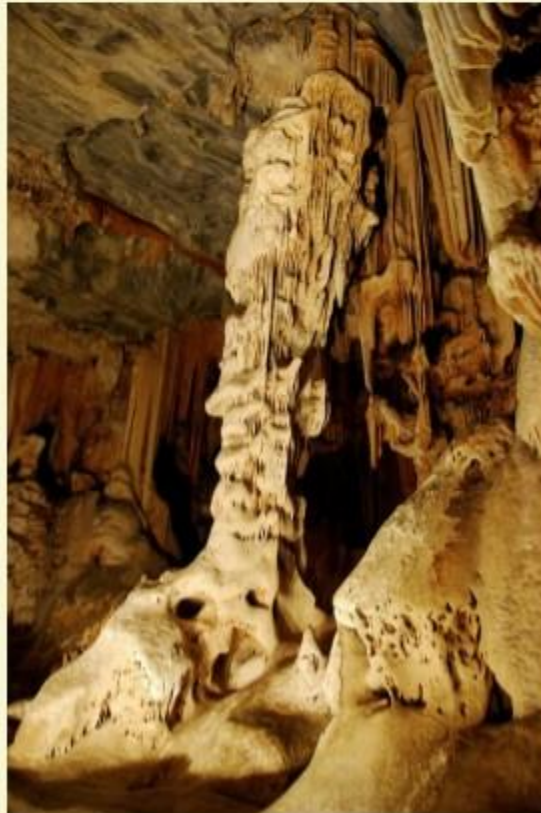
3 Stalagmites may take the shape of a column, a disc with either a smooth, rounded bulging end or a miniature crater like depression.

KARST TOPOGRAPHY

DEPOSITIONAL LANDFORMS

STALACTITES, STALAGMITES AND PILLARS

- 4 The stalagmite and stalactites eventually fuse to give rise to columns and pillars of different diameters.



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